

Research Synthesis & Upgrade Plan

AI-Driven Marp/Markdown Presentation Workflow

Agenda

Today's Topics

1. Design Best Practices

Mayer's principles, typography, layout

2. Color, Accessibility & Data Viz

WCAG, palettes, chart frameworks

3. Narrative & Anti-Patterns

SCQA, slide count, F1-F8

4. AI Tool Landscape

6 tools surveyed, key gap found

5. Open-Source Pipelines

sunbigfly, Akxan, DeepSlide

6. Vision QA & Upgrade Roadmap

VLM contracts, 4-phase plan

Design Best Practices

- **Cognitive Load Theory** – Mayer's 12 principles, validated across 200+ experiments
- **Coherence** – every element must serve the message, nothing extraneous
- **Signaling** – use contrast and bold for key takeaways, never font size alone
- **Segmenting** – max 5 +/- 2 items per slide, progressive disclosure
- **Pre-training** – define key terminology upfront in title and outline



Typography & Visual Hierarchy

- **7-level scale:** 48pt > 40pt > 28pt > 22pt > 18pt > 14pt > 12pt
- **Line height:** body 1.5x, headings 1.1x (unitless CSS)
- **Hard rules:** body $\geq 14\text{px}$ and $\leq 24\text{px}$, max 3 font families
- **7 bento templates:** full-bleed, 2-col, 3-col, 4-card, 6-card, center



Color, Accessibility & Data Viz

- **Palette structure:** Base > Surface > Primary > Accent > Text (3 levels)
- **WCAG AA:** 4.5:1 body, 3:1 large text – avoid red-green pairs
- **Max 5 colors per slide**, decorative gradients excluded
- **Best charts:** bar (comparison), line (trends), scatter (correlation)
- **Avoid:** pie charts over 3 segments, 3D charts, dense tables



Narrative Structure & Failure Modes

- **SCQA**: Situation > Complication > Question > Answer (McKinsey)
- **Pacing**: 1 slide per 2 minutes, 15 min talk = 7-8 slides
- **F2 Overfill** – content overflow, reduce count
- **F6 Contrast failure** – below WCAG threshold
- **F8 Narrative disconnect** – no story arc, restructure with SCQA



AI Presentation Tool Landscape

Tool	Layout Engine	Quality Mechanism
Gamma.app	Template-constrained	Human review, no visual QA
Beautiful.ai	Constraint satisfaction	Template design rules only
Tome	Narrative-first	No pixel-level QA
Canva Magic	Score heuristics	Layout balance ranking
MS Copilot	Template-constrained	Template guardrails
Pitch	Design system	Human review only

Key gap: No major tool uses vision feedback to refine slide appearance

Open-Source Agentic Pipelines



sunbigfly/ppt-agent-skills

Multi-agent state machine, 4 subagents, JSON contract architecture, Pillow pixel QA, stateless resume



Akxan/ppt-agent-skill

26 production styles, 14 typography rules, 18 chart types (SVG), 8 failure modes with fix protocols



DeepSlide (arXiv)

Time-budgeted planner, Markov style inheritance, dual-scoreboard evaluation (artifact + delivery)

Common Gaps & Best Format

VLM Gap

No open-source tool uses vision-language models for semantic slide QA. Analysis is purely pixel-statistical.

BEST FORMAT

HTML/CSS as primary target, convert to Marp markdown

DEEPSLIDE INNOVATION

Logical-chain planner with time budgets, style inheritance, dual-scoreboard evaluation

Vision QA & Style Contracts

- **Stage A:** per-slide visual audit – balance, readability, color
- **Stage B:** deck-wide narrative audit – arc, progression, pacing
- Use VLM for **semantic** checks, CSS/DOM for **quantitative**
- **theme-contract.json:** full design system encoding per deck



Upgrade Roadmap

Phase 1 – Foundation

Style contract system,
CSS/DOM inspection QA,
density contract validator

WEEK 1

Phase 2 – Pipeline

5-stage pipeline shell, per-
slide JSON contracts,
stateless resume support

WEEK 2

Phase 3 – Quality

VLM QA pass with structured
prompts, feedback loop,
chart MCP tool

WEEK 3

Phase 4 – Polish

Icon search MCP tool,
narrative arc planning,
integration testing

WEEK 4